

# Lesson 11 Regularization

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April 16 (Wednesday), 2025

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- Linear Regression (Reminder)

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- Elastic Net Regression

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- When we have a lot of observations we can be fairly confident that the Least Squares line accurately reflects the relationship between  $y$  and  $x$ .

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- $\lambda \geq 0$  is the **tuning** parameter

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- There is no need to shrink the intercept, which is simply a measure of the mean value of the response when  $x_{i1} = x_{i2} = \dots = x_{ip} = 0$ .

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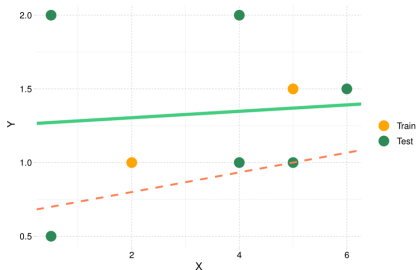
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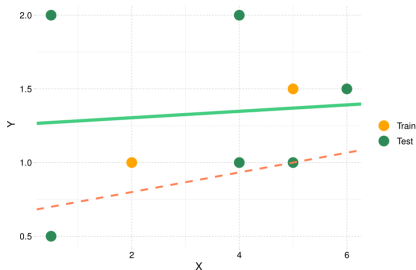
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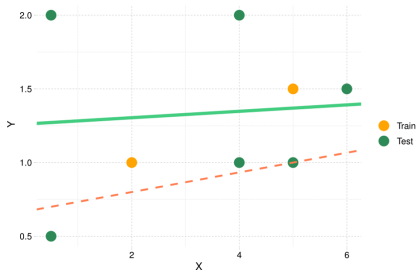
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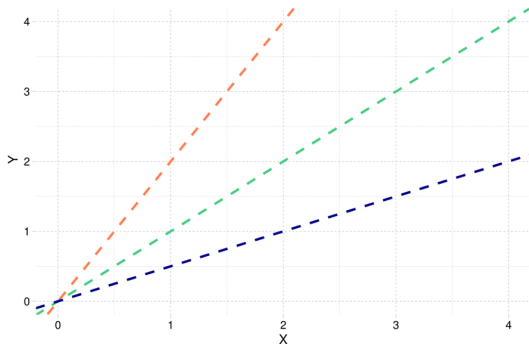
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- Shrinking the coefficient estimates can significantly reduce their variance.

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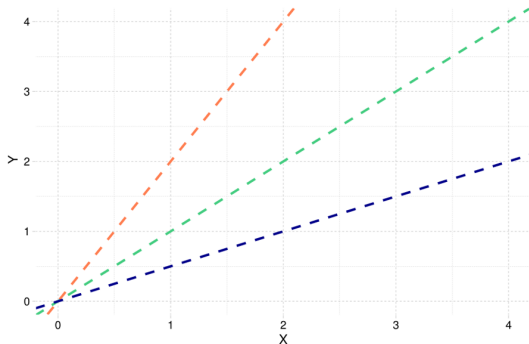
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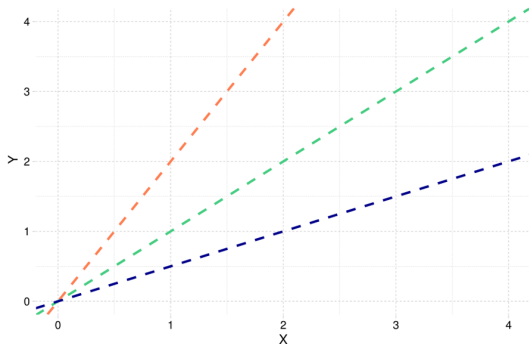
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- $\lambda \rightarrow \infty, \hat{\beta}_{ridge} \rightarrow 0$
- Less and less sensitive to  $x$  variable

# Decrease in the slope: example

April 15 (Tuesday), 2025

- Suppose we have  $y_i = \beta x_i + \varepsilon_i$  and we have the following data:

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- Suppose  $\lambda = 240$
- $\hat{\beta}_{ridge} \frac{90}{6+240} = \frac{90}{6+246} = 0.37 < 15$

# Solving the problem of non-existing coefficient

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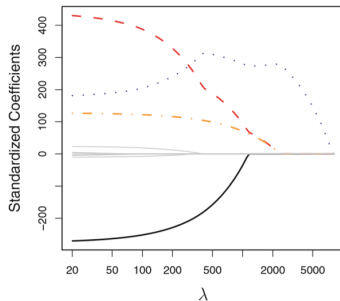
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- $\hat{\beta}_{ridge} = \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2 + \lambda} = \frac{90}{0 + \lambda}$
- $\text{rank}(X) < m \Rightarrow (X^T X)^{-1}$  does **NOT** exist, but  $(X^T X + \lambda I)^{-1}$  does exist.

# Coefficient as a function of lambda

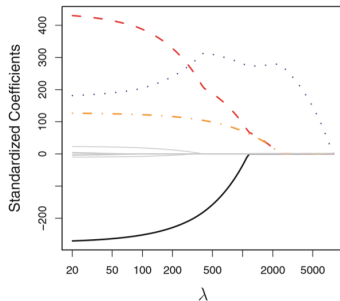
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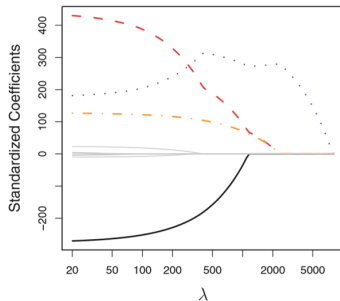
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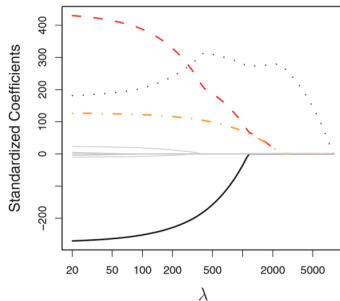
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- The graphs are taken from Introduction to Statistical Learning

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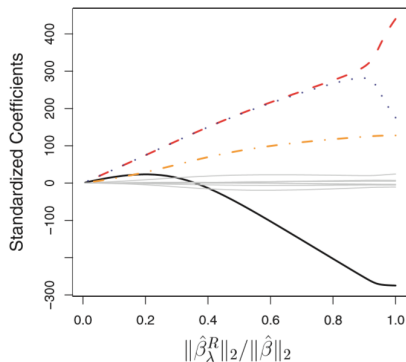
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- The amount that the ridge regression coefficient estimates have been shrunk towards zero is the 2 norm of the ridge regression coefficient estimates divided by the 2 norm of the least squares estimates.



# Norms and measurement



- $\lambda \uparrow \Rightarrow \|\hat{\beta}_{ridge}(\lambda)\|_2 \downarrow \Rightarrow \|\hat{\beta}_{ridge}(\lambda)\|_2 / \|\hat{\beta}\|_2 \downarrow$

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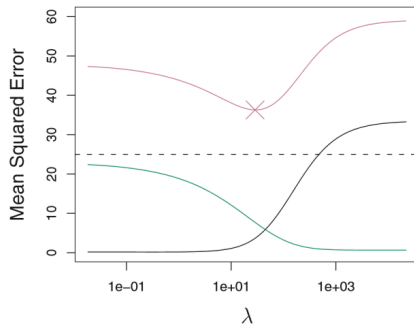
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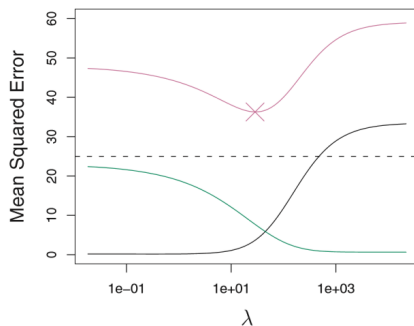
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# Bias-variance trade-off as a function of lambda



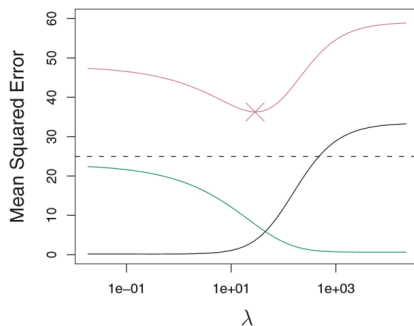
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- The **MSE** drops considerably as  $\lambda$  increases from 0 to 10.



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- Lasso Regression is similar to Ridge Regression with the difference in Penalty term.

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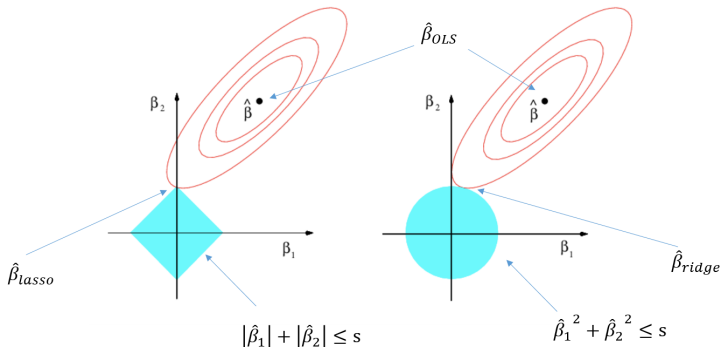
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- If  $s$  is sufficiently large, then the constraint regions will contain  $\hat{\beta}$ , and so the ridge regression and lasso estimates will be the same as the least squares estimates.

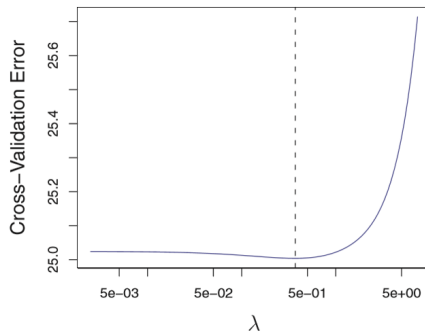
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- Since ridge regression has a circular constraint with no sharp points, this intersection will not generally occur on an axis, and so the ridge regression coefficient estimates will be exclusively non-zero.

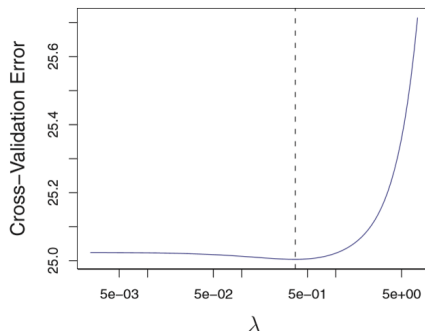
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- Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

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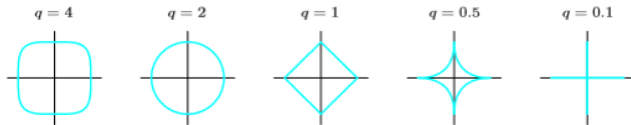
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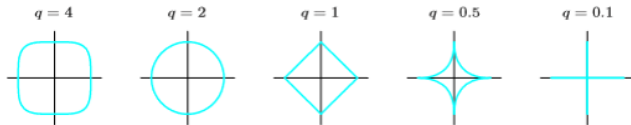
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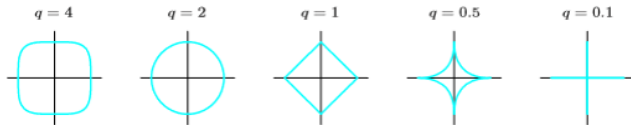
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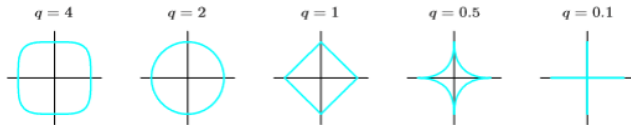
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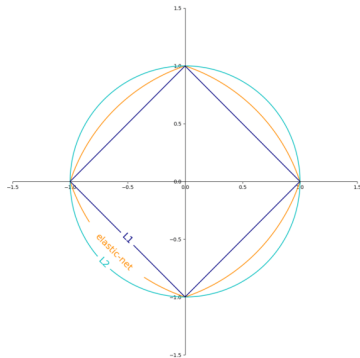
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- The most popular combination is **Elastic Net Regression**.

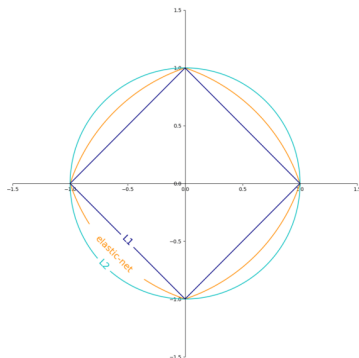
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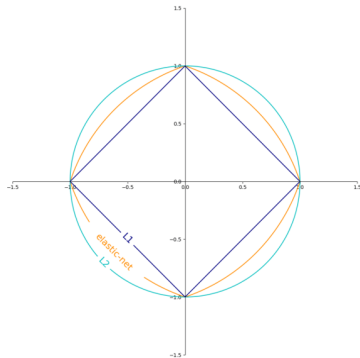
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